Fundamentals of Circuits and Electronics

CSAI 205

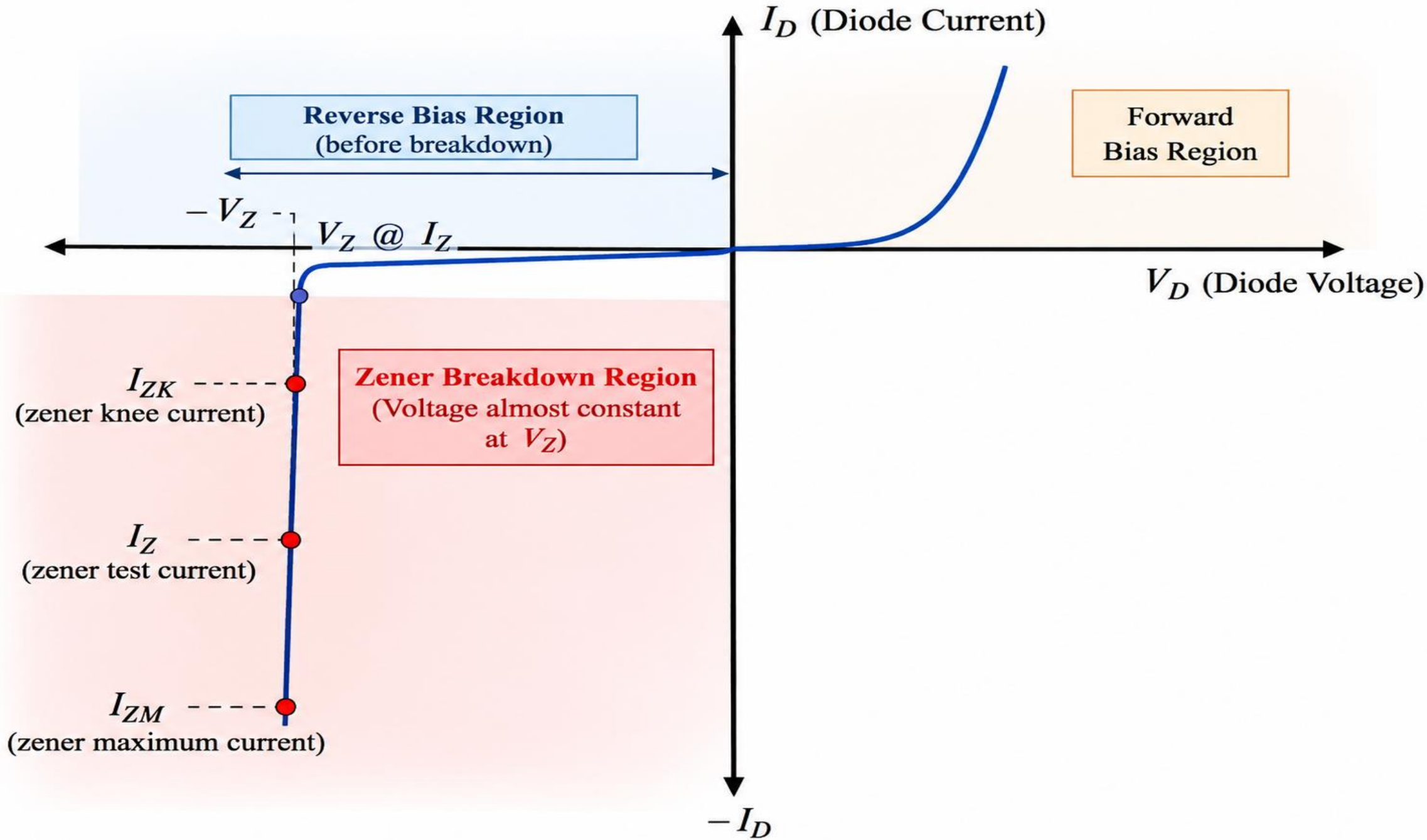
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- **Zener diode**
- Optical diodes:
  - **Light-emitting diodes (LEDs)**
  - **Photodiodes**
  - Laser Diode
  - Optocouplers
- Schottky diode
- **Varactor diode**
- PIN diode
- Tunnel diode

Zener Diodes

- Zener diodes are designed to operate in **reverse breakdown**. Zener breakdown occurs in a zener diode **at low reverse voltages (5 up to 15 V)**. A zener diode is heavily doped to reduce the breakdown voltage. This causes a very thin depletion region.
- As a result, an intense electric field exists within the depletion region. Near the zener breakdown voltage (V_Z), the field is intense enough to pull electrons from their valence bands and create current.




I_D (Diode Current)
Reverse Bias Region
(before breakdown)
Forward
Bias Region
$-V_Z$
V_Z @ I_Z
V_D (Diode Voltage)
I_{ZK}
(zener knee current)
Zener Breakdown Region
(Voltage almost constant
at V_Z)
I_Z
(zener test current)
I_{ZM}
(zener maximum current)
$-I_D$

Compute zener voltage at current (a) I = 30 mA, (b) I = 10 mA according to the following zener data sheet: (Z_{ZT} = 5 Ω, V_{ZT} = 6.8 V @ I_{ZT} = 20 mA, I_{ZK} = 1 mA, I_{ZM} = 50 mA)

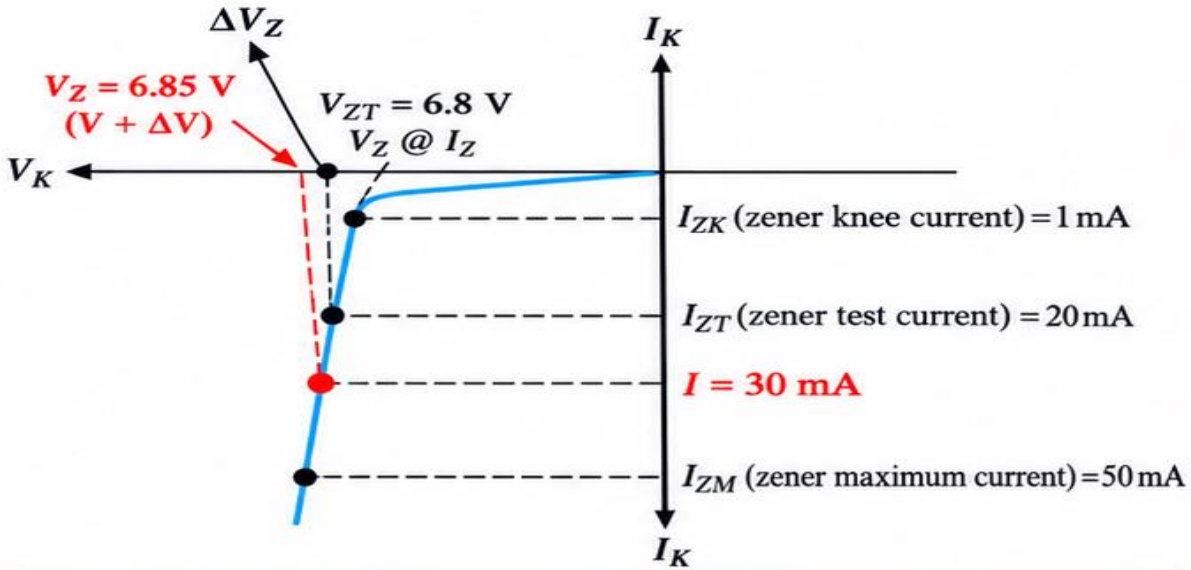
(a) At I = 30 mA (apply on the graph)

Solution:

$V_Z = V_{ZT} + \Delta V_Z$

But $Z_{ZT} = \Delta V / \Delta I$

$V_Z = V_{ZT} + (Z_{ZT} \cdot \Delta I) = 6.8 + [5 \times (30 - 20) \times 10^{-3}] = 6.85$ V



Similarly, At I = 10 mA (apply on the graph)

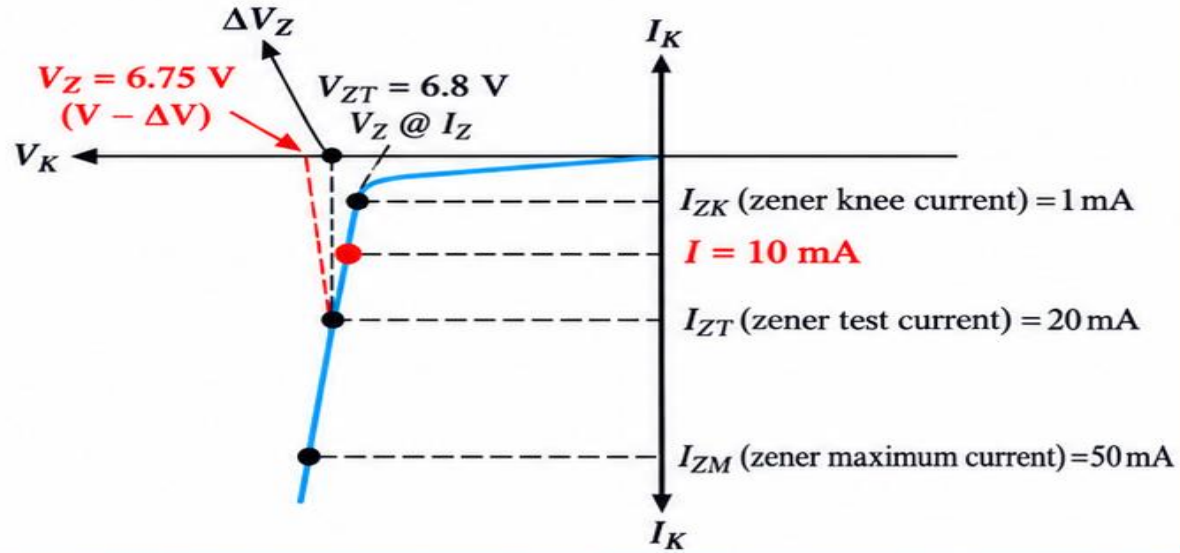
Solution:

We will find the required voltage below test value

$V_Z = V_{ZT} - \Delta V_Z$

But $Z_{ZT} = \Delta V / \Delta I$

$V_Z = V_{ZT} - (Z_{ZT} \cdot \Delta I) = 6.8 - [5 \times (20 - 10) \times 10^{-3}] = 6.75$ V



Summary of results:

I_Z (mA)	V_Z (V)
10	6.75
30	6.85

Conclusion: current range have a great change However, voltage range remains constant

هذا لمثال يوضح اهمية وجود (zener) في الدوائر حيث بالرغم من ان التغير في التيار كبير لكن الجهد تقريبا ظل ثابتا

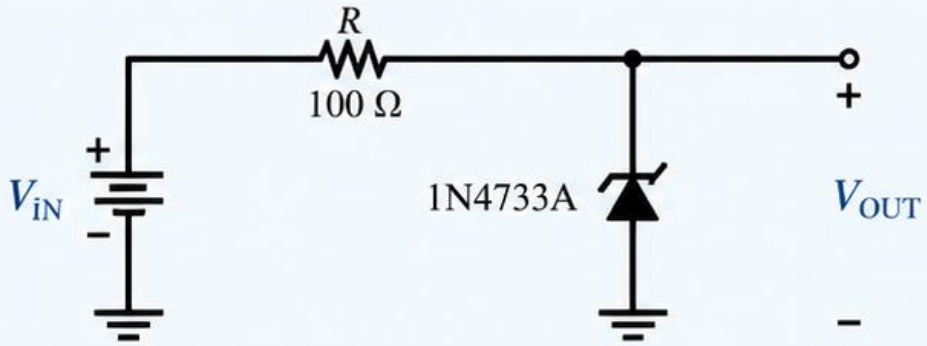
Zener diode Applications

- **Zener as voltage regulator**
- **Zener as simple limiter or clipper.**

a. Zener as a voltage regulator

Example

Determine the minimum and the maximum input voltages that can be regulated by the zener diode in the shown figure.
Where $V_Z = 5.1$ V at $I_{ZT} = 49$ mA, $I_{ZK} = 1$ mA, $I_{ZM} = 196$ mA and $Z_Z = 7\ \Omega$ at I_{ZT}.



Solution

(a) To get minimum V_{in}

We will get the minimum input voltage at I_{ZK} and in this case V_{out} will be minimum.

$$V_{out}(min) = V_Z - \Delta V_Z$$
$$\Delta V_Z = (I_{ZT} - I_{ZK}) \times Z_Z$$
$$\Delta V_Z = (49 - 1) \times 7 = 0.336\ V$$
$$V_{out}(min) = 5.1 - 0.336 = 4.764$$
$$V_{in}(min) = V_{out}(min) + I_{ZK} \,.\, R = 4.764 + 1(0.1) = 4.864\ V$$

(b) To get maximum V_{in}

We will get the maximum input voltage at I_{ZM} in this case V_{out} is maximum.

$$V_{out}(max) = V_Z + \Delta V_Z$$
$$\Delta V_Z = (I_{ZM} - I_{ZT}) \times Z_Z$$
$$\Delta V_Z = (196 - 49) \times 7 = 1.03\ V$$
$$V_{out}(max) = 5.1 + 1.03 = 6.13$$
$$V_{in}(max) = V_{out}(max) + I_{ZM} \,.\, R = 6.13 + 196(0.1) = 25.7\ V$$

Summary of results

I_Z (mA)	V_{out} (v) = V_Z	V_{in} (v)
1	4.764	4.864
196	6.13	25.7

Line regulation

$$\text{Line regulation} = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{6.13 - 4.764}{25.7 - 4.864} = \textit{very small value} = \textit{Excellent regulation}$$

Find the minimum load resistance (Ω) for the shown network.

1 Relate load voltage, current and minimum resistance

$$V_{RL} = I_{Lmax} \cdot R_{min}$$

$$or, \quad R_{min} = \frac{V_{RL}}{I_{Lmax}} = \frac{5}{I_{Lmax}} \quad --------(1)$$

2 Find the loop current using KVL

Find the loop current I using KVL around the source–resistor–zener loop:

$$10 - IR - 5 = 0$$

$$10 - 100I - 5 = 0$$

$$100I = 5$$

$$I = 0.05\ A = 50\ mA$$

This is the total current entering the node before it splits into zener current and load current.

3 Current splitting at the node

$$I = I_{Zk} + I_{Lmax}$$

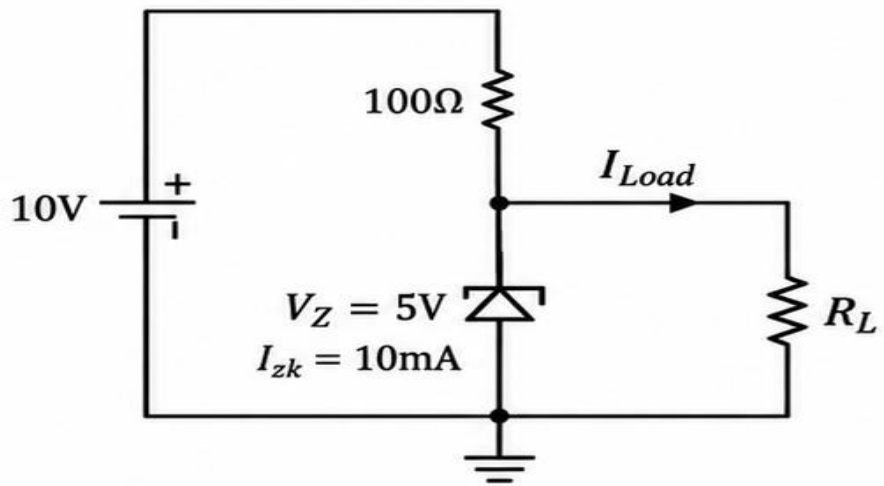
$$or, \quad I_{Lmax} = I - I_{Zk} = 50 - 10 = 40\ mA$$

4 Minimum load resistance

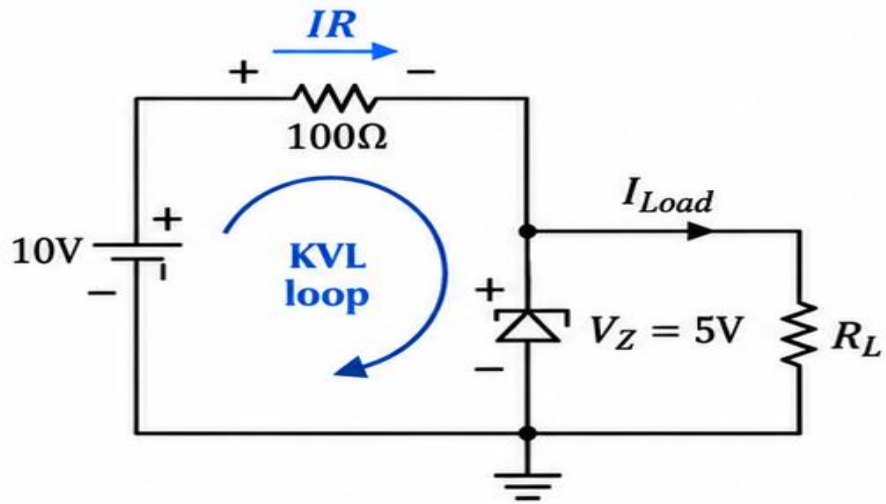
$$R_{min} = \frac{5}{40 \times 10^{-3}} = \frac{5000}{40} = \frac{500}{4} = 125\ \Omega$$

Minimum load resistance, $R_{min} = 125\ \Omega$

Given network

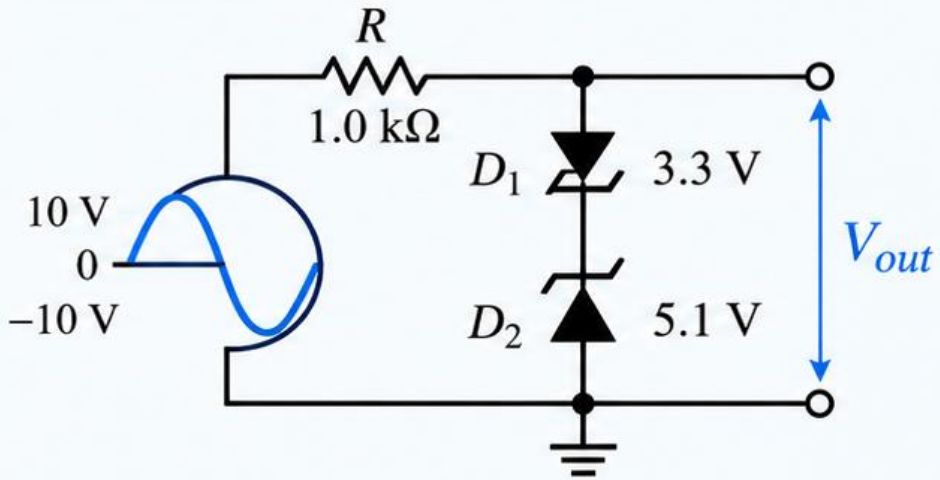


KVL loop and reverse-bias polarity

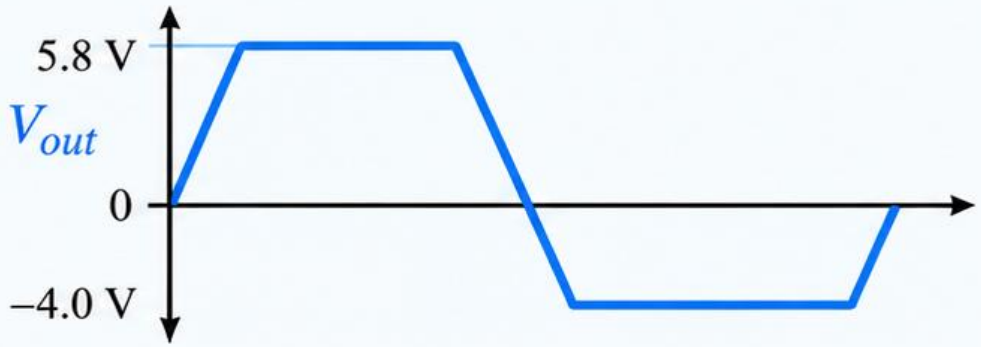


b. Zener as a clipper or limiter

Circuit Diagram



Output Waveform (V_{out})



(+ve half cycle)

- D_1 (3.3 V) is forward biased and represented by 0.7 V
- D_2 (5.1 V) is reverse biased (open circuit), the zener action is $V_Z = 5.1$ V then ($V_{out} = V_{in}$).
- When V_{in} exceed 5.1 V, then D_2 will get into breakdown ($V_{out} = 5.1 + 0.7 = 5.8$).

(−ve half cycle)

- D_1 is reverse (open circuit) till $V_{in} = -3.3$ then ($V_{out} = V_{in}$).
- D_2 is represented by 0.7 V
- When V_{in} exceed −3.3 then D_1 will get into breakdown ($V_{out} = -3.3 - 0.7 = -4$).

Embedded Systems
Part (2)

1. LDR (Light Dependent Resistor)

A photoresistor changes resistance when light changes.

Concept

The LDR has high resistance in darkness and lower resistance in bright light.

It is usually used in a voltage divider with a fixed resistor.

Arduino reads the divider voltage at an analog pin, such as A0.

Example application

Automatic night lamp

Smart streetlight

Simple light meter

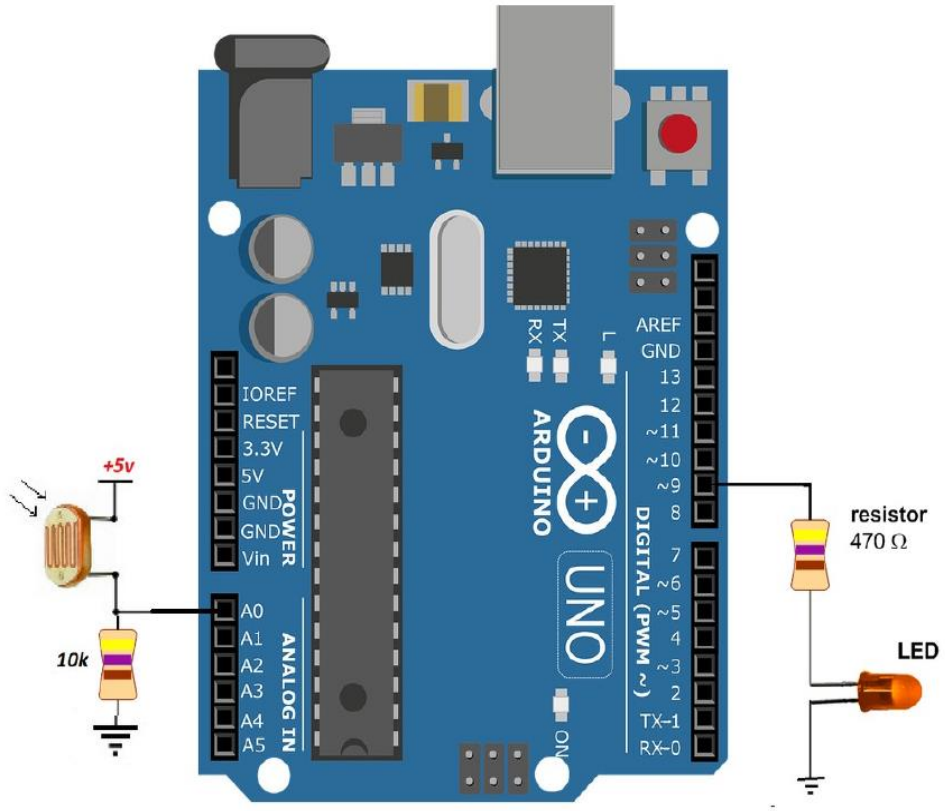
Key example / notes

Example: if the room becomes dark, the Arduino can turn on an LED automatically.

The threshold value must be tested in the real environment.

(1) LDR circuit and code

The wiring and the code use A0 for the LDR divider and D9 for the LED.




```
const int ldrPin = A0;
const int ledPin = 9;
const int threshold = 200;

void setup() {
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
}

void loop() {
  int lightValue = analogRead(ldrPin);
  if (lightValue <= threshold) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
  }
  Serial.println(lightValue);
  delay(200);
}
```

2. Pulse Width Modulation (PWM)

PWM gives analog-like control using a digital output.

Concept

A PWM pin rapidly switches between HIGH and LOW.

The average effect depends on the duty cycle.

On Arduino Uno, common PWM pins include 3, 5, 6, 9, 10, and 11.

Example application

LED brightness control

Motor speed control

Audio tone intensity

Key example / notes

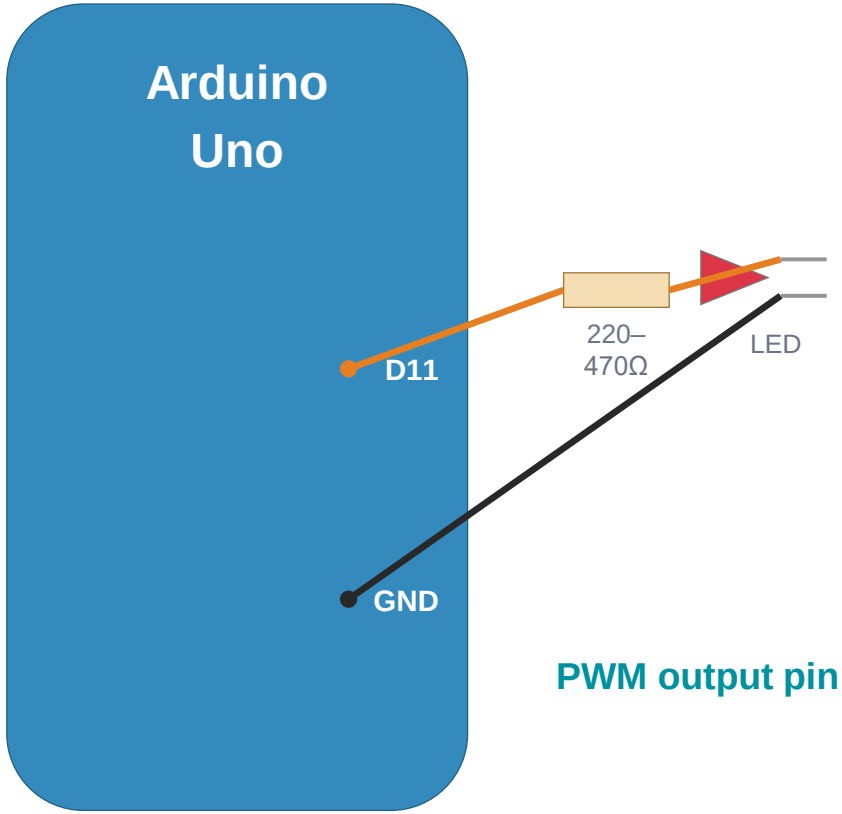
Example: analogWrite(64) gives a dim LED, analogWrite(128) gives medium brightness, and analogWrite(255) gives maximum brightness.

PWM is excellent for fading LEDs.

PWM LED circuit and code

Pin D11 is used as the PWM output to fade the LED.

Circuit diagram



PWM output pin: D11

```
const int ledPin = 11;

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  for (int fade = 0; fade <= 255; fade += 3) {
    analogWrite(ledPin, fade);
    delay(20);
  }
  for (int fade = 255; fade >= 0; fade -= 3) {
    analogWrite(ledPin, fade);
    delay(20);
  }
}
```

3. LM35 Temperature Sensor

LM35 outputs a voltage that is proportional to temperature.

Concept

LM35 is an analog sensor.

Its scale factor is approximately 10 mV per °C.

Arduino converts the output voltage into a digital value using analogRead().

Example application

Room temperature monitor

Weather station

Heat warning system

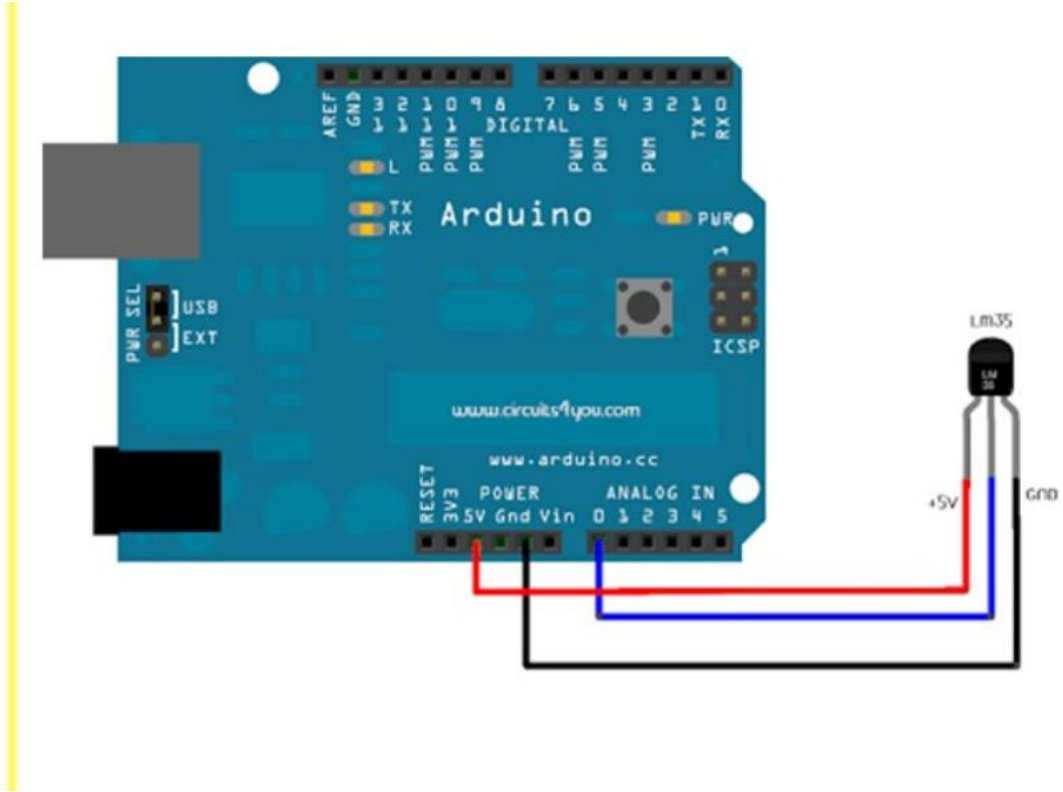
Key example / notes

Example: 25 °C produces about 0.25 V at the output pin.

If the temperature rises, the output voltage rises as well.

LM35 circuit and code

LM35 VOUT is connected to A0; power and ground must be correct.




```
const int lm35Pin = A0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  int raw = analogRead(lm35Pin);
  float volts = raw * (5.0 / 1023.0);
  float celsius = volts * 100.0;
  Serial.print("Temperature: ");
  Serial.print(celsius, 1);
  Serial.println(" C");
  delay(1000);
}
```

LM35 Temperature Conversion (Example: 25°C)

1. **analogRead(lm35Pin)**: Arduino's 10-bit ADC converts the LM35 analog voltage (0.25 V) into a digital value: raw ≈ 51 (range: 0–1023).
2. **raw * (5.0 / 1023.0)**: Converts the digital reading back into voltage, assuming a 5 V reference:
   volts = 51 × 5 / 1023 ≈ 0.249 V
3. **volts * 100.0**: Converts voltage into Celsius because LM35 produces 10 mV (0.01 V) per 1°C:
   celsius = 0.249 × 100 ≈ 24.9°C

4. Rain Sensor

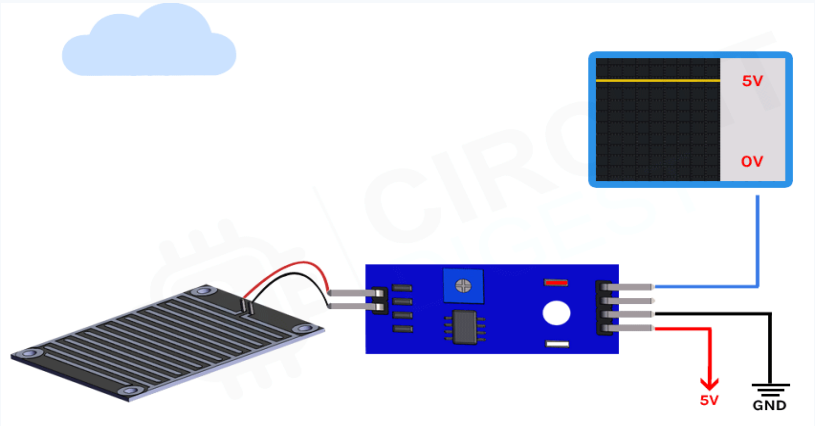
The rain sensor detects water drops using a conductive sensing plate.

Concept

The sensing plate changes conductivity when it becomes wet.

The module may provide analog output (AO) and digital output (DO).

In this circuit, AO is used and read by an analog pin.



Example application

Rain alarm

Automatic wiper

Smart irrigation system

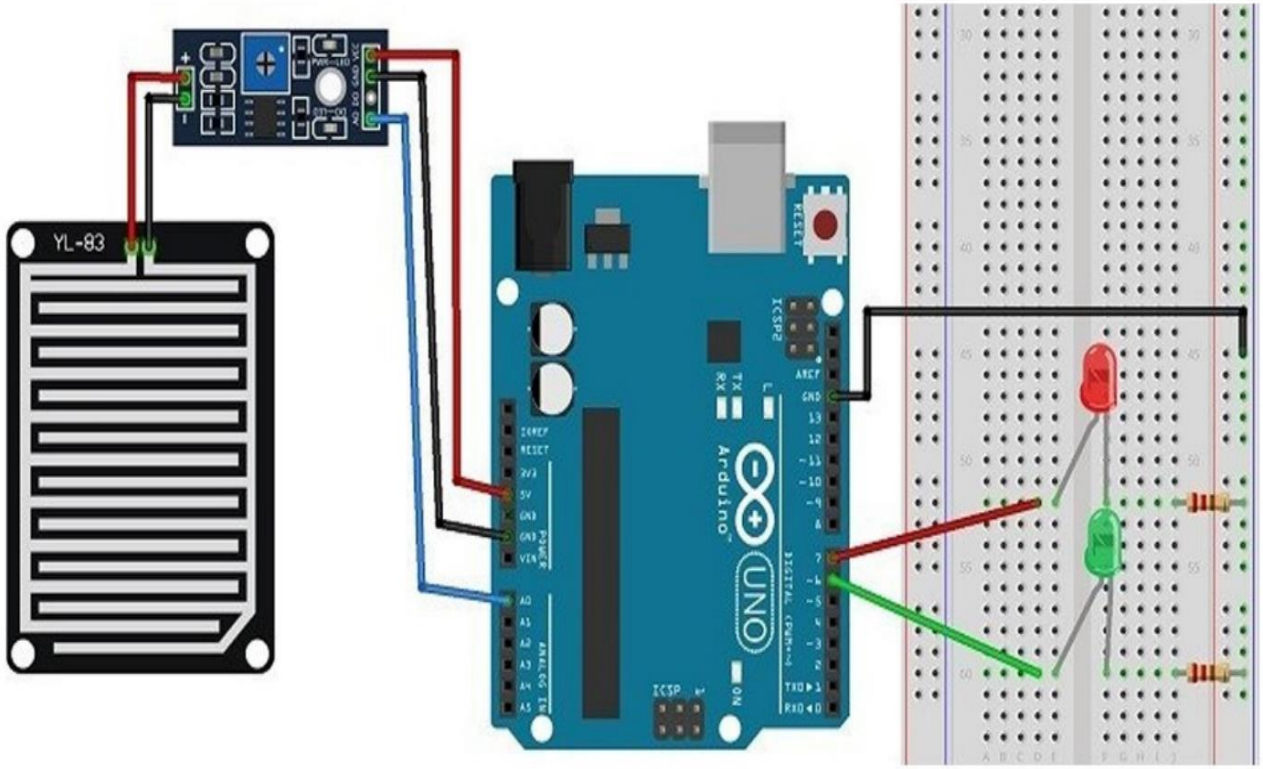
Key example / notes

Example: a dry plate gives one reading range and a wet plate gives another range.

Use Serial Monitor to compare dry, damp, and wet values.

Rain sensor circuit and code

This wiring reads the analog output AO at pin A0.




```
const int rainPin = A0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  int reading = analogRead(rainPin);
  Serial.print("Rain sensor AO: ");
  Serial.println(reading);
  delay(500);
}
```

5. Gas Sensor

A gas sensor changes its output according to gas concentration.

Concept

Many gas modules provide an analog output that Arduino can read.

The sensor usually needs warm-up time before stable readings.

A threshold can be used for a simple demonstration alarm.

Example application

Kitchen gas warning

Air quality experiment

Smoke detection prototype

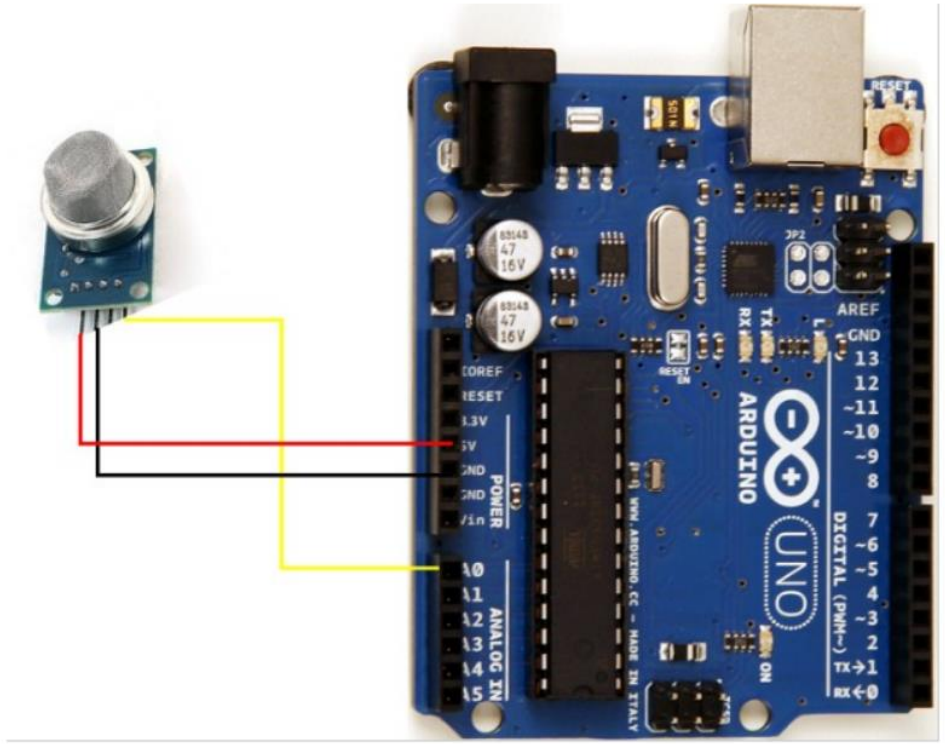
Key example / notes

Example: if the ADC reading rises above a selected threshold, the system can print a warning message.

Real calibration depends on the gas type and sensor model.

Gas sensor circuit and code

The analog output is connected to A0.




```
const int gasPin = A0;
const int threshold = 300;

void setup() {
  Serial.begin(9600);
}

void loop() {
  int value = analogRead(gasPin);
  Serial.print("Gas ADC: ");
  Serial.println(value);
  if (value > threshold) Serial.println("Gas high");
  else Serial.println("Gas normal");
  delay(500);
}
```

6. Ultrasonic Sensor (HC-SR04)

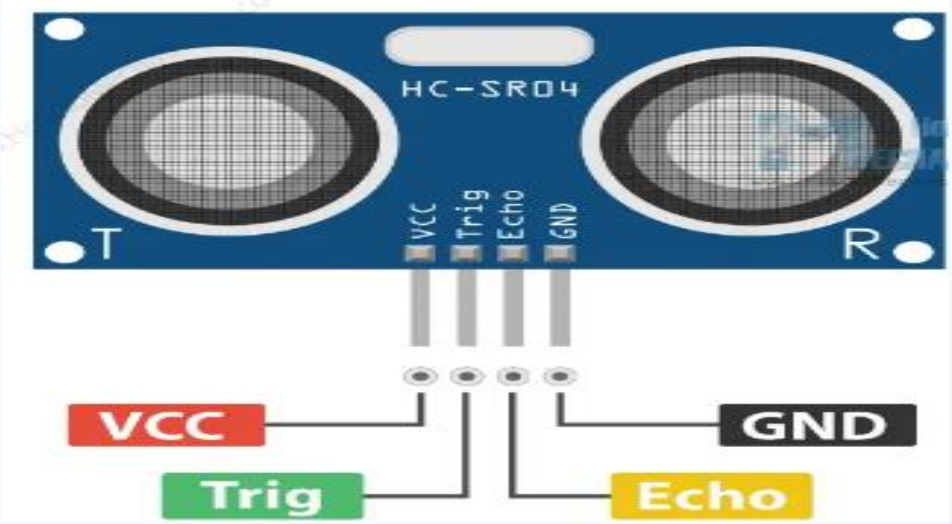
Ultrasonic distance sensing uses sound waves and echo timing.

Concept

The Arduino sends a short pulse to the Trig pin.

The sensor emits ultrasound and listens for the reflected wave.

The Echo pin stays HIGH for a time proportional to the distance.



Example application

Parking assistant

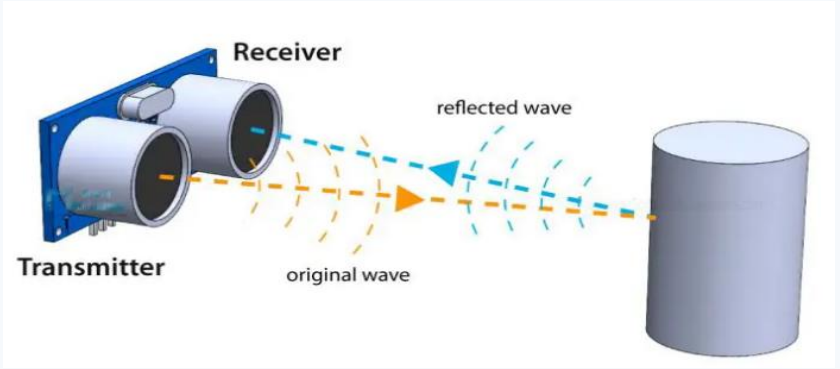
Obstacle detection

Liquid level measurement

Key example / notes

Example: if an object is closer, the echo returns sooner.

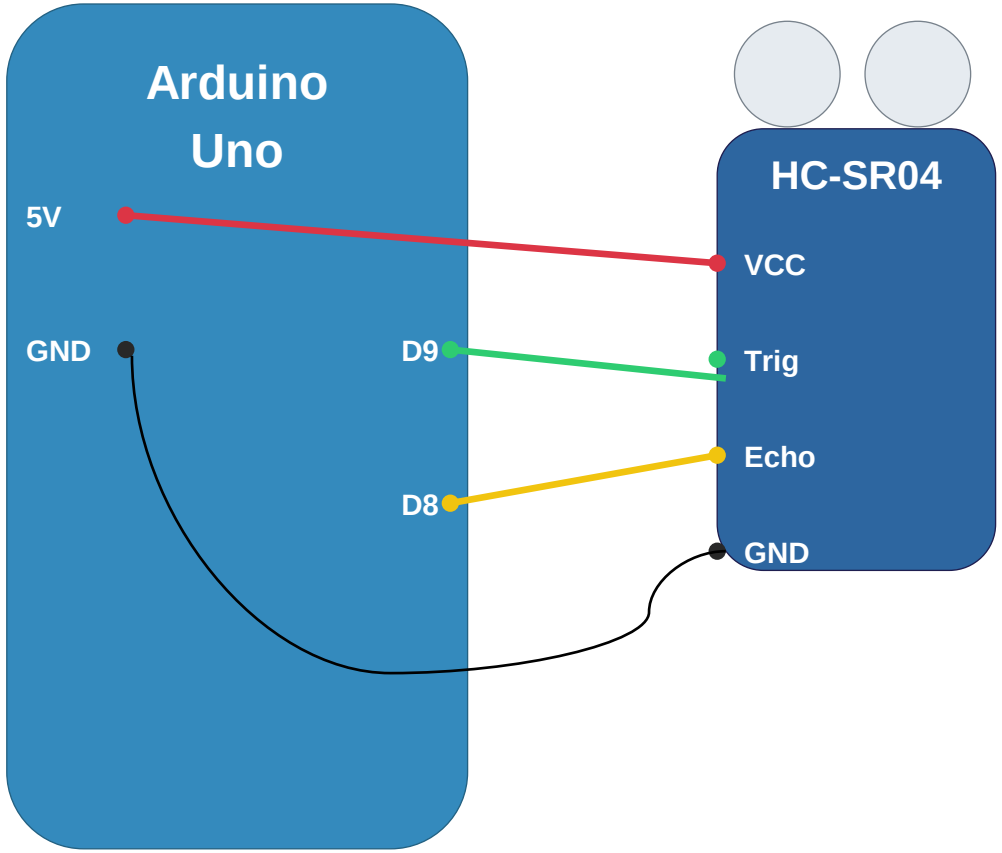
Distance = speed of sound × time / 2.



Ultrasonic circuit and code

Trig is connected to D9 and Echo is connected to D8.

Circuit diagram




```
const int trigPin = 9;
const int echoPin = 8;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  unsigned long t = pulseIn(echoPin, HIGH);
  float d = t * 0.0343 / 2.0;
  Serial.println(d);
  delay(200);
```

7. IR Obstacle Sensor

An infrared sensor detects nearby objects from reflected IR light.

Concept

The module contains an IR emitter and an IR receiver.

When an object reflects the IR beam, the output changes state.

Many modules are **active-LOW**, which means LOW indicates detection.

Example application

Line-following robot

Obstacle avoidance robot

Automatic object-triggered alarm

Key example / notes

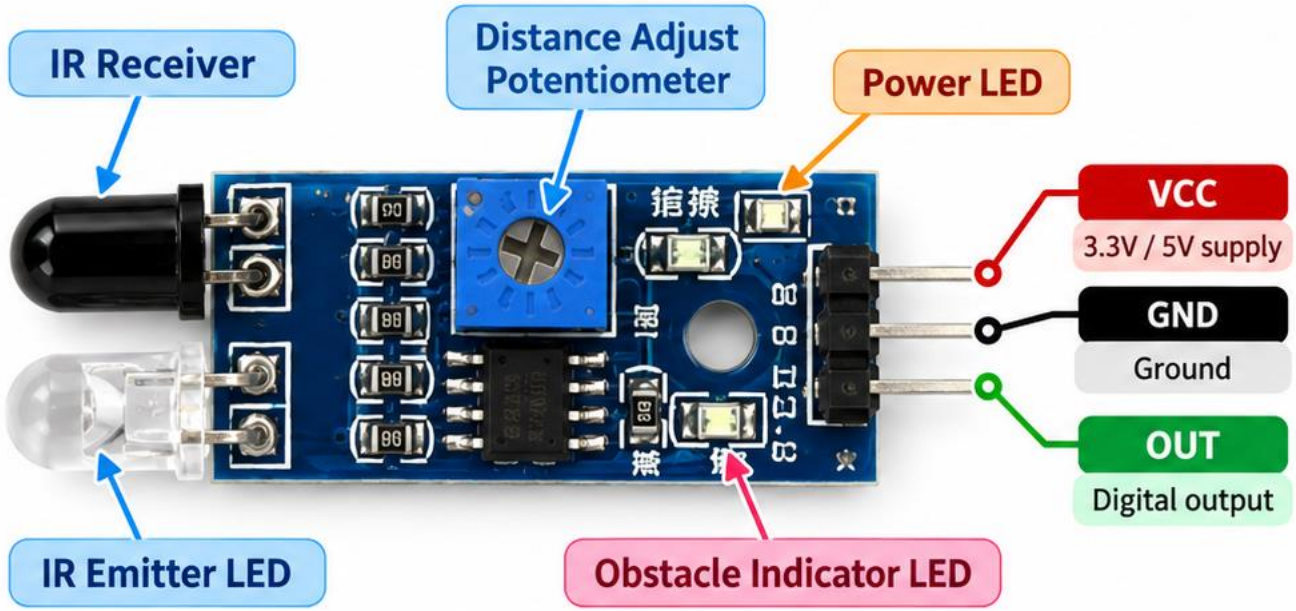
Example: a robot can stop or beep when an object appears in front of the sensor.

The sensing distance is usually adjusted with the onboard potentiometer.

7) IR Sensor (Infrared Obstacle Sensor Module)

How it works

1. The IR LED emits infrared light.
2. If an object is in front of the sensor, the reflected IR light is received.
3. The module comparator changes the OUT pin state when the detection threshold is reached.



What the potentiometer does

- It adjusts the detection sensitivity / distance.

Typical uses

- Obstacle detection
- Line-following / robotics
- Simple object sensing

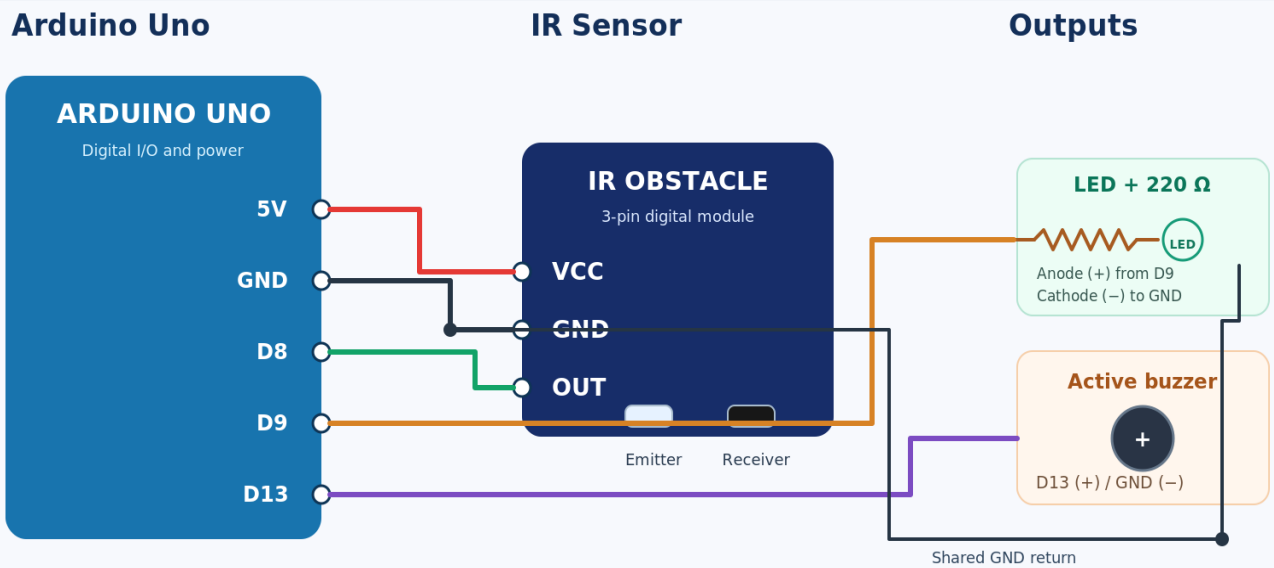
Student notes

- Works from 3.3 V or 5 V depending on the module.
- OUT is a digital signal for microcontrollers such as Arduino or ESP boards.

IR Sensor — Circuit Diagram & Arduino Code

CIRCUIT · PIN-TO-PIN WIRING

IR Obstacle Sensor — Circuit Wiring According to Code



VCC → 5V **GND → GND** **OUT → D8** **LED → D9** **BUZZER → D13**

CODE · MATCHES THE CONNECTIONS

```
const int sensorPin = 8;
const int outputPin = 9;
const int buzzerPin = 13;

void setup() {
  pinMode(sensorPin, INPUT);
  pinMode(outputPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
}

void loop() {
  int sensorValue = digitalRead(sensorPin);
  if (sensorValue == LOW) {
    digitalWrite(outputPin, HIGH);
    digitalWrite(buzzerPin, HIGH);
  } else {
    digitalWrite(outputPin, LOW);
    digitalWrite(buzzerPin, LOW);
  }
}
```

IR OUT = LOW → LED ON + buzzer ON

8. Servo Motor

A servo motor moves to a commanded angular position.

Concept

A standard servo has three wires: power, ground, and signal.

The signal is a timed control pulse, not ordinary LED PWM fading.

Arduino uses the Servo library to command the angle.

Example application

Robot arm

Door or gate control

Camera pan mechanism

Key example / notes

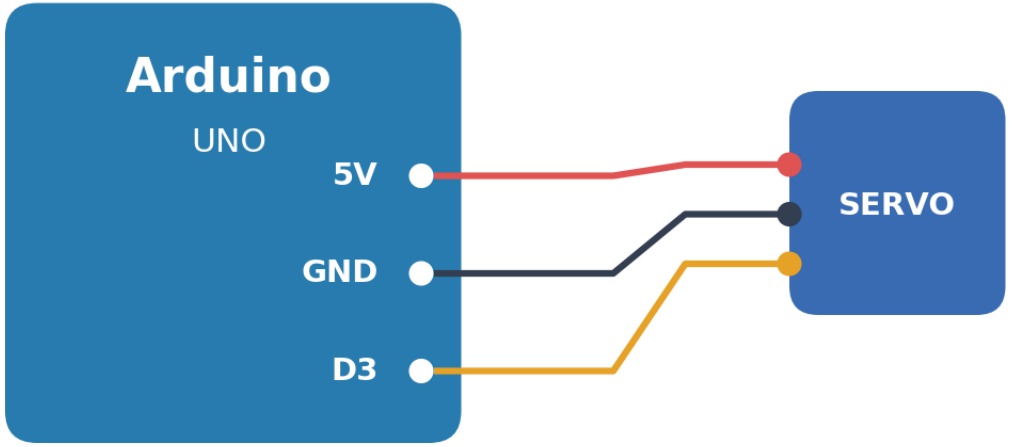
Example: write(0), write(90), and write(180) move the shaft to three positions.

The servo should have a stable power supply.

8. Servo Motor | Circuit & Arduino Code

A servo moves to a commanded angular position using timed control pulses.

CIRCUIT DIAGRAM



5V → VCC • GND → GND • D3 → SIGNAL

ARDUINO CODE

```
#include <Servo.h>
Servo myServo;
const int servoPin = 3;

void setup() {
  myServo.attach(servoPin);
}

void loop() {
  myServo.write(0);
  delay(2000);
  myServo.write(90);
  delay(2000);
  myServo.write(180);
  delay(2000);
}
```

KEY CONCEPT **write(0) → 0° | write(90) → 90° | write(180) → 180°**

Stable supply • common ground • servo position is controlled by pulse timing, not ordinary LED PWM.